Configurable Static Type Checking for Forth

Evaluator and Examples

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EuroForth 2026

Static Type Checking

- Detect stack errors before execution
- Detect type mismatches early
- Precisely document word behaviour

Research Evolution

- 1990 - 1998: Stack Effect Calculus
- 2002 - 2006: Typed Wildcards and Inheritance
- 2008: Typing Tools for Typeless Stack Languages (Java prototype)
- 2026: Configurable Static Type Checking for Forth – Type checker in gforth

From an informal comment to a checkable contract

```
: SQUARED  ( n -- n² )  DUP * ;
```

INFORMAL SEMANTIC COMMENT

n² says what the result means mathematically.

- Useful documentation
- May be absent or stale
- Consumed as source text; not trusted as an assertion

CHECK

TYPED SYMBOLIC EFFECTS

```
DUP  ( X[1] -- X[1] X[1] )
*    ( n n  -- n )
```

Inferred:

```
SQUARED  ( n -- n )
```

The checker verifies stack balance, cell types and cell identity, not the mathematical claim that the result equals n².

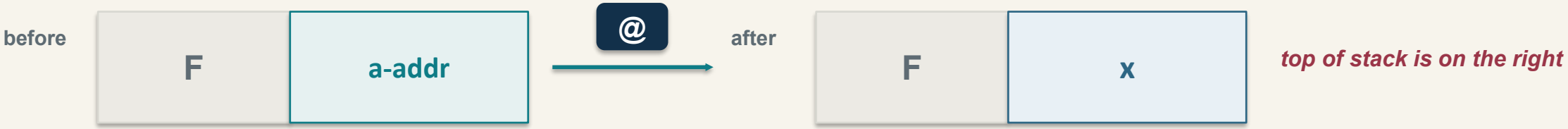
Stack effect

required input suffix

`@ ( a-addr -- x )`

guaranteed output suffix

Any deeper frame F passes through unchanged



top of stack is on the right

An effect is a typed symbolic pre/post contract over only the data-stack suffix (top right) a phrase inspects.

A boundary can refine or fail

COMPATIBLE

DP C@ 0=

DP (-- a-addr)

C@ (c-addr -- char)

0= (n -- flag)

result: (-- flag)

The trace retains a-addr because it is more specific than c-addr.

CLASH

C@ !

C@ (c-addr -- char)

! (X a-addr --)

char ✖ **a-addr**

produced required on top

The checker rejects the incompatible boundary before the source can attempt the bad store.

Types express the project's compatibility policy

CONVENIENT FORTH-LIKE PROFILE

more specific → more general

a-addr < c-addr < addr < X

char < n < X

flag < n < X

Useful when the target treats flags and characters as numeric cells.

STRICTER STANDARD-LIKE PROFILE

the selected file makes these choices explicit

a-addr < c-addr < addr < u < n < x

char < +n < u < n < x

flag < x (separate from n)

Aliases may name one canonical type: X, x, and cell can be equivalent.

Composition keeps the more specific comparable boundary type; incomparable types clash.

Indices remember which cell is which

DUP

(X[1] -- X[1] X[1])

one input, two correlated outputs

SWAP

(X[2] X[1] –
X[1] X[2])

two origins exchange positions

OVER

(X[2] X[1] –
X[2] X[1] X[2])

the deeper input reappears on top

same index = same abstract origin • **different indices** do not claim unequal concrete values • **no index** = fresh symbolic item

Indices are local logical variables not addresses or runtime tags. The checker freshens them for every word occurrence.

Three operations used for control structures

SEQUENCE

$p \cdot q$

Boundary composition

Can q consume what p leaves? Match from the top inward and propagate the more precise type through every correlated occurrence.

ALTERNATIVES

$p \sqcap q$

Common effect

Attempt to reconcile both path effects into one interface. This is a single must-style syntactic summary—not a set of every possible behavior.

REPETITION

$p \sqcap (p \cdot p)$

One-versus-two

Compare one pass with two consecutive passes. Accept only when the implemented merge finds a locally stable summary.

The untyped base has a polycyclic-monoid structure; the checker adds partial, subtype-aware, identity-sensitive matching.

Loops receive a local stability check not an invariant proof

```
: ULOOP   BEGIN DUP 0= UNTIL   ;
```

IMPLEMENTED ONE-VERSUS-TWO TEST

```
p  =  DUP 0= UNTIL
```


```
ULOOP  ( n[1] -- n[1] )
```

WHAT THIS RESULT SAYS

- **Useful:** detects many bodies that grow, shrink, or change incompatibly on a second pass.
- **Local only:** the duplicated copy feeds 0= and UNTIL; the original n remains.
- **Not proved:** termination, every iteration count, or a general inductive loop invariant.

Treat the reported loop effect as a finite stack-stability approximation, not as full loop verification.

Parsing and control are part of the specification

WORD ENTRIES: SOURCE ROLE + STACK EFFECT

```
"("  parse until ")"        ( -- )
:    parse word define colon   ( -- )
;    control end               ( -- )
literal integer               ( -- n )
```

Metadata says how the reader consumes delimiters, names, definitions, and control boundaries.

The effect models runtime data-stack behavior separately from compile-time source handling.

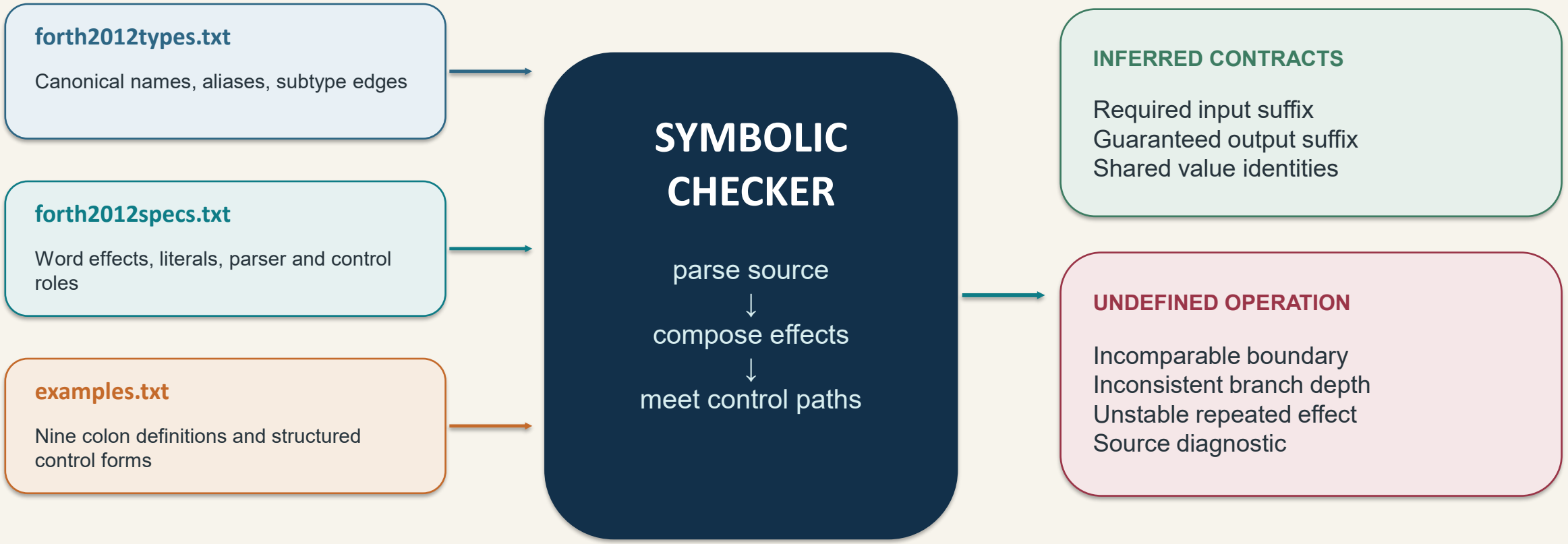
CONTROL SYNTAX + EFFECT RECIPE

```
syntax:
  IF <then branch>
  [ELSE <else branch>] FI
effect:
  IF
  either <then branch> <else branch>
```

The syntax block captures regions; the effect block combines them by sequence, either, or repeat.

The profile can rename FI to THEN or change the type hierarchy without changing the effect algorithms.

Three inputs produce one inferred dictionary



No concrete values are executed. At ; the inferred body effect becomes the specification of the new word.

Example runs

forth2012types.txt + forth2012specs.txt + examples.txt

NINE COLON DEFINITIONS

Each group isolates one operation.

test1–3 straight-line composition
test4–6 conditional branch meet
test7–9 repetition and type variance

INFER

THE QUESTIONS

sequence + subtyping
identity propagation
branch contract meet
finite repetition

Every inferred effect shown here is the exact examples.txt.log result; comments are not used as declarations.

Concepts covered

The examples are deliberately small: each makes one part of the effect calculus visible.

test1–2

Composition

Produced and required suffixes match from the top inward; fresh results carry types without identities.

test3

Identity flow

A late consumer refines an earlier input backward through SWAP and DUP.

test4–6

Branch contracts

Implicit prefixes align unequal displays; only common type and identity guarantees survive.

test7–8

Repetition

One pass meets two passes to obtain a finite local stability summary.

test9

Variance

Inputs retain the common subtype; outputs retain the common supertype.

: test1 (-- value) 3 4 5 - + ;

: test1	empty	
3 (-- n)	n	3
4 (-- n)	n n	3 4
5 (-- n)	n n n	3 4 5
- (n n -- n)	n n	3 -1
+ (n n -- n)	n	2
;	Results in (-- n)	

```
: test2  ( fl1 fl2 -- FALSE flor )
    OR FALSE SWAP ;
```

```
: test2                                  flag flag
OR ( flag flag -- flag  )                flag
FALSE ( -- flag )                        flag flag
SWAP  ( X[2] X[1] – X[1] X[2] )          flag flag
;                                        Results in ( flag flag -- flag flag )
```

```
: test3  ( addr X -- X addr value )
    SWAP DUP @ ;
```

```
: test3                                  a-addr[2] X[1]
SWAP ( X[2] X[1] -- X[1] X[2] )          X[1] a-addr[2]
DUP ( X[1] -- X[1] X[1] )                X[1] a-addr[2] a-addr[2]
@  ( a-addr -- X )                       X[1] a-addr[2]  X
;                                        Results in
                                         ( a-addr[2] X[1] -- X[1] a-addr[2] X )
```

```
: test4  (  n incr -- n1  )
    IF 1 + FI ;
```

```
: test4                    n flag
IF (flag -- )              n
1 ( -- n )                 n n
+  ( n n -- n )            n
FI                         Branch effect ( n -- n )
;                          Results in ( n flag -- n )
```

If-branch body effect (n – n) is more exact than empty branch effect (--)

```
: test5  ( X1 flag -- X1 X2 )
    IF DUP ELSE 0 FI ;
```

: test5	X[1] flag
IF (flag --)	X[1]
DUP (X[1] -- X[1] X[1])	X[1] X[1]
ELSE	If branch effect (X1 -- X1 X1)
0 (-- n)	X[1] n
FI	Else-branch effect (-- n)
;	Meet: (X1 -- X1 X2)

Prefer supertype in output (we don't know which branch was chosen and n is subtype of X)
Results in (X[1] flag -- X[1] X)

```
: test6  ( X2 X1 ad fl -- X3 X4 X5 )
    IF ROT ELSE @ FI ;
```

```
: test6                                          X2  X1  a-addr  flag
IF ( flag -- )                                   X2  X1  a-addr
ROT ( X[3] X[2] X[1] -- X[2] X[1] X[3]  )        X1  a-addr  X2
ELSE                                             If ( X2 X1 a-addr -- X1 a-addr X2 )
@ ( a-addr -- X )                                X2  X1  X3
FI                                               Else ( a-addr -- X )
;                                                Meet:  ( X2 X1 a-addr -- X3 X4 X5 )
```

Prefer supertype in output (we don't know which branch was chosen, a-addr is subtype of X)
Results in (x x a-addr flag -- x x x)

```
: test9  ( ad fl -- X )
    IF @ ELSE C@ FI ;
```

```
: test9
IF ( flag -- )
@ ( a-addr -- X  )
ELSE
C@ ( c-addr -- char )
FI
;
```

```
 a-addr  flag
a-addr
X
If ( a-addr -- X )
char
Else ( c-addr -- char )
Meet:  ( a-addr -- X )
```

Prefer subtype in input (a-addr is subtype of c-addr and X is supertype of char)
Results in (a-addr flag -- X)

```
: test7  ( fl2  fl1 --  fl3  fl4  )
    BEGIN SWAP OVER WHILE INVERT REPEAT ;
```

```
: test7
BEGIN
SWAP ( X[2] X[1] -- X[1] X[2] )
OVER ( X[2] X[1] -- X[2] X[1] X[2] )
WHILE
INVERT  ( flag -- flag )
REPEAT
;
```

fl2 fl1

Loop1 effect: (X X -- X X)

fl1 fl2

fl1 fl2 fl1

fl1 fl2

fl1 fl3

Loop2 effect: (flag -- flag)

Prefer subtype in input (flag is subtype of X)
Results in (flag flag -- flag flag)

```
: test8  (  --  )
   BEGIN KEY 0= UNTIL ;
```

```
: test8
BEGIN
KEY (  -- char )
0= ( n -- flag )
UNTIL ( flag -- )
;
```

char

flag

Loop body (--)

Prefer subtype in input (char is subtype of n)
Results in (--)

What acceptance means and what it does not

ACCEPTANCE SUPPORTS

- Declared word effects compose without a modeled clash
- Every checked branch receives one implemented common effect
- Every checked loop passes the local one-versus-two test
- Source structure, parser roles, and compilation state fit the active profile

Useful as executable stack-effect documentation.

ACCEPTANCE DOES NOT PROVE

- That trusted primitive and parser declarations match runtime behavior
- A path-sensitive set of every branch behavior
- Termination or an inductive invariant for arbitrary loop counts
- Unrestricted EXECUTE, return-stack tricks, non-local THROW, or metaprogramming

Use alongside tests, review, and target-system knowledge.

Known limits include lossy identity compaction, one effect per primitive, and the one-versus-two loop approximation.

Summary

types + specifications + source → inferred specifications / alarms

1. **Composition is sequential** — A late boundary refinement propagates through every occurrence of the same symbolic value. Type information flows back and forth in a linear sequence of words.

2. **Branch meet is conservative** — Inputs retain the common subtype; outputs retain the common supertype; identities must hold on every path.

3. **Repetition is deliberately finite** — One-versus-two gives a useful local summary—not termination or a general loop invariant.

Configurable, executable stack-effect documentation with explicit and bounded guarantees.